Tricky Topic

Expanding double brackets

You need to be able to expand the **product** of two brackets. Use one of these three methods.

1 One-at-a-time

You can expand one bracket first, then the other:

$n(n + 3)$

$(n + 3)(n + 8) = n(n + 3) + 8(n + 3)$
$= n^2 + 3n + 8n + 24$
$= n^2 + 11n + 24$

$+ 8(n + 3)$

Now expand and simplify as normal.

Remember to collect like terms.

2 FOIL

The acronym **FOIL** tells you the four terms you need to find:

$2a^2$ $-b^2$

$(2a + b)(a - b) = 2a^2 - 2ab + ab - b^2$
$= 2a^2 - ab - b^2$

ab

$-2ab$

First terms
Outer terms
Inner terms
Last terms

Some people remember this as a 'smiley face'.

3 Grid method

You can draw a grid to help you expand the brackets:

$(x + 7)(x - 5) = x^2 - 5x + 7x - 35$
$= x^2 + 2x - 35$

Remember to collect like terms if possible.

	x	-5
x	x^2	$-5x$
7	$7x$	-35

The negative sign belongs to the 5. You need to write it in your grid.

Worked example

Expand and simplify $(y + 2)(y + 9)$ **(2 marks)**

$(y + 2)(y + 9) = y^2 + 9y + 2y + 18$
$= y^2 + 11y + 18$

y^2 18

$(y + 2)(y + 9)$

$2y$

$9y$

This example shows the **FOIL** method. The question says expand and **simplify** so collect like terms at the end.

Worked example

Expand and simplify $(3p - 4)^2$ **(2 marks)**

$(3p - 4)^2 = (3p - 4)(3p - 4)$
$= 9p^2 - 12p - 12p + 16$
$= 9p^2 - 24p + 16$

	$3p$	-4
$3p$	$9p^2$	$-12p$
-4	$-12p$	16

When you square a term in brackets you multiply the **whole bracket** by itself. Start by writing out:
$(3p - 4)^2 = (3p - 4)(3p - 4)$
This example shows the **grid method** but you could use any method to expand the brackets.
Remember that $3p \times 3p = 9p^2$ and be careful with the **negative signs**:
$-4 \times -4 = 16$ $p \times -4 = -4p$

Now try this

1 Complete this working:
$(x + 6)(x + 3) = x^2 + 3x + \square x + \square$
$= x^2 + \square x + \square$ **(2 marks)**

2 Expand and simplify
(a) $(x + 5)(x - 1)$ **(2 marks)**
(b) $(p - 6)^2$ **(2 marks)**

Worked solution video

Quadratic graphs

Quadratic equations contain an x^2 term. Quadratic equations have **curved** graphs. You can draw the graph of a quadratic equation by completing a table of values.

Worked example

(a) Complete the table of values for $y = 4x - x^2$ **(2 marks)**

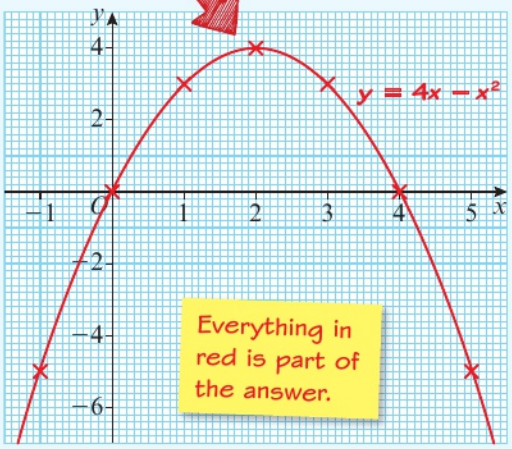
x	−1	0	1	2	3	4	5
y	−5	0	3	4	3	0	−5

(b) On the grid, draw the graph of $y = 4x - x^2$ **(2 marks)**

(c) Write down the coordinates of the turning point. **(1 mark)**

(2, 4)

The **turning point** is the point where the direction of the curve changes.



When $x = -1$: $4 \times (-1) - (-1)^2$
$= -4 - 1 = -5$
When $x = 4$: $4 \times 4 - 4^2 = 16 - 16 = 0$
Plot your points carefully on the graph and join them with a **smooth** curve.

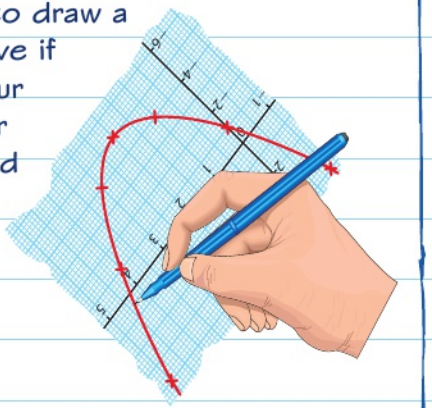
Check it!
All the points on your graph should lie on the curve. If one of the points doesn't fit then double check your calculation.

Drawing quadratic curves

- Use a sharp pencil.
- Plot the points carefully.
- Draw a smooth curve that passes through every point.
- Label your graph.
- Shape of graph will be either ∪ or ∩

Drawing a smooth curve

It's easier to draw a smooth curve if you turn your graph paper so your hand is **inside** the curve.



Now try this

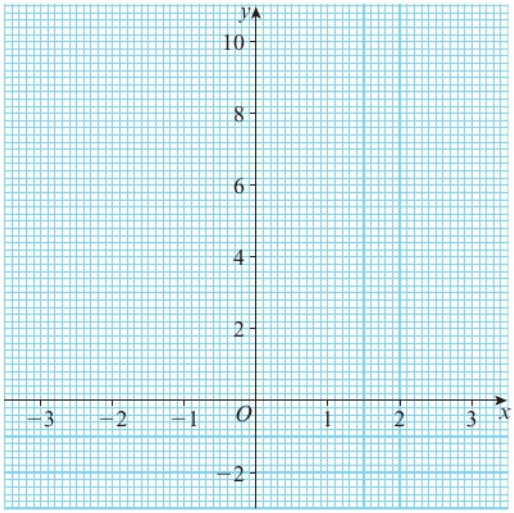
Worked solution video

(a) Complete this table of values for $y = x^2 + 2$ **(2 marks)**

x	−3	−2	−1	0	1	2	3
y		6		2	3	6	

(b) Draw the graph of $y = x^2 + 2$ for $x = -3$ to $x = 3$ **(2 marks)**

(c) Use your graph to find the value of y when $x = 2.5$ **(1 mark)**



Using quadratic graphs

You might need to read values off a quadratic graph. You can do this by looking at the points where the graph crosses the x-axis, or by drawing a horizontal line on your graph.

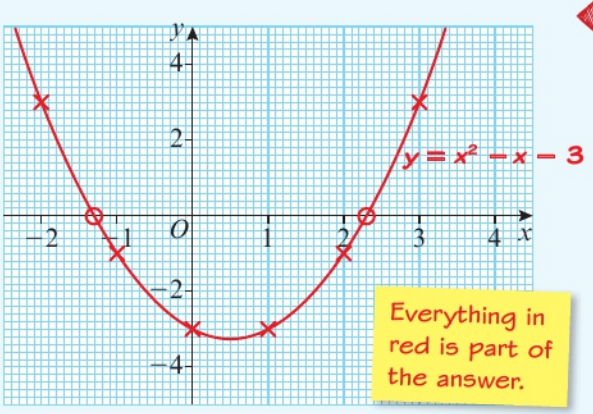
Worked example

(a) Complete the table of values for $y = x^2 - x - 3$ **(2 marks)**

x	−2	−1	0	1	2	3
y	3	−1	−3	−3	−1	3

This is an example of a **quadratic graph**. Its equation contains an x^2 term. You should always use a table of values to draw quadratic graphs.

(b) On the grid, draw the graph of $y = x^2 - x - 3$ **(2 marks)**



Everything in red is part of the answer.

Examiners' report

Don't use a ruler – you need to join your points with a **smooth curve**.

Real students have struggled with questions like this in recent exams – **be prepared!**

(c) Use your graph to write down the solutions of $x^2 - x - 3 = 0$ **(2 marks)**

$x = -1.3$ and $x = 2.3$

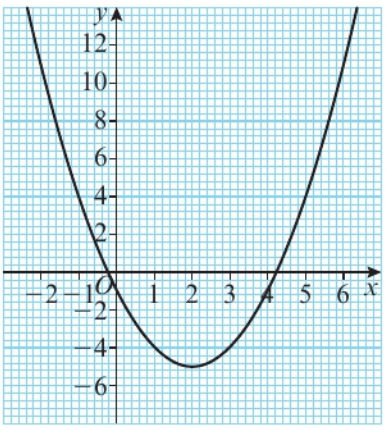
The solutions of the equation $x^2 - x - 3 = 0$ are the x-coordinates at the points where the graph crosses the x-axis. These are sometimes called the **roots** of the equation. You should always read graphs accurate to the nearest small square. So on this graph your answers will be accurate to 1 decimal place.

Now try this

This is the graph of $y = x^2 - 4x - 1$

(a) Write down the values of x where the graph crosses the x-axis. **(2 marks)**

(b) Write down the values of x when $y = 8$ **(2 marks)**



Draw a horizontal line on the graph at $y = 8$.
Read down to the x-axis at the points where the line crosses the curve.
Read the graph correct to the nearest small square.
These values of x are the solutions to the quadratic equation $x^2 - 4x - 1 = 8$